How Matrix Size Affects the Runtime and Memory of Naive Matrix Multiplication in Python, Java and C

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Repository: https://github.com/JoelLopezGuillen/big-data-matrix-multiplication (folder `assignment-1/`, tag `assignment-1`)

Abstract

This study measures how the matrix size *n* affects the runtime and peak memory of the same naive i-j-k triple-loop dense matrix multiplication implemented in C, Java and Python on a single laptop (Intel Core i7-12650H, Windows 11). All implementations read identical float64 input matrices, were validated against NumPy, and were timed over five repetitions after warm-up runs, reporting the median and interquartile range. Runtime did not follow a single cubic curve: the time per inner-loop iteration stayed roughly constant while the strided operand fitted in one cache level and rose sharply each time it exceeded one (in C: 0.27, 0.5, 2.0 and 13.3 ns per iteration). C and Java performed similarly from n = 256 to 512, while CPython was 36–112 times slower than C at the sizes where both completed. The largest feasible sizes within a 900 s per-process budget were 2048 for C and 1024 for Java and Python; a supplementary run showed that the Java timeout at n = 2048 was caused by the budget design, not by infeasibility. These conclusions apply only to the tested implementations, settings, inputs and machine.

1. Introduction and objective

Dense matrix multiplication is a standard benchmark kernel because its arithmetic work grows as $O(n^3)$ while its data grows as $O(n^2)$, so it stresses both the processor and the memory hierarchy [3], [11]. Implementing the same algorithm in languages with different execution models (ahead-of-time compiled C, JIT-compiled Java and interpreted CPython) makes it possible to observe how execution model and memory layout affect performance under controlled conditions.

Research question. How does the matrix size *n* affect the runtime and peak memory of the same naive triple-loop dense matrix multiplication implemented in Python, Java and C, on one fixed machine?

Hypotheses (stated before the main results were analysed).

- **H1.** Runtime grows approximately as n^3, i.e. by a factor of about 8 each time n doubles.
- **H2.** C and Java, which execute native machine code, are much faster than CPython, which interprets bytecode.
- **H3.** In Python, interpreter overhead dominates each operation, so memory-hierarchy effects should be smaller than in C and Java and growth should stay closer to ×8.

2. Background

Naive matrix multiplication. For C = A·B with n × n matrices, each element C[i][j] is the dot product of row *i* of A and column *j* of B. The classic i-j-k triple loop performs n^3 multiply-add operations, so doubling n multiplies the arithmetic work by $2^3 = 8$.

Row-major storage and access pattern. In row-major order, consecutive elements of a row are adjacent in memory, and element (i, k) of a flat array is at index i·n + k. In the i-j-k order the inner loop reads A[i][k] contiguously but reads B[k][j] with a stride of n elements (8n bytes), touching a different cache line on every iteration.

Memory hierarchy. Processors move data between memory levels in cache lines (64 bytes on x86). Each level is larger and slower than the previous one: L1, L2, L3 and main memory [3], [4]. When the data a loop reuses no longer fits in a level, accesses are served by the next, slower level. On the test machine each performance core has a 48 KB L1 data cache and a 1.25 MB L2 cache, and the 24 MB L3 cache is shared [2], [12].

Execution models. C is compiled ahead of time by gcc to native code. Java is compiled to bytecode that the HotSpot JVM first interprets and then compiles just-in-time to optimised native code for frequently executed ("hot") methods [5]; measurements taken before this happens are not representative, which is why warm-up runs are needed [10]. CPython interprets bytecode and represents every number as a heap-allocated object, so each arithmetic operation involves type checks and object management.

Benchmarking pitfalls. Measurements must use a monotonic clock, exclude warm-up runs, and consume the result so that the compiler cannot remove the computation as dead code [1], [10].

3. Methodology

3.1 Implementations

All three implementations use the same algorithm: the i-j-k triple loop, accumulating the dot product in a local variable and then writing `C[i][j] = sum`. No optimised matrix multiplication library is used in any implementation.

- **C and Java** store each matrix as a flat row-major `double` array of length n^2, indexed as `i*n + k`.
- **Python** uses a list of lists of `float`, the idiomatic representation in pure Python. This is an unavoidable language-specific difference: each element is an 8-byte pointer to a separate float object (about 24 bytes), instead of an 8-byte value stored contiguously.
- The Python version stores the current row of A in a local variable (`a_row = a[i]`) outside the inner loops. gcc performs an equivalent optimisation automatically at `-O2`.
- The output matrix is allocated and zero-initialised inside the multiplication function (`calloc` in C, `new double[]` in Java, a list comprehension in Python).

3.2 Inputs

Input matrices A and B were generated once per size with NumPy's `default_rng` (PCG64), using seed 42 for A and 43 for B, with values drawn uniformly from [-1, 1) as float64. They were stored as little-endian, row-major binary files that all three implementations read. A shared file was used instead of three language-specific random generators, which produce different sequences, so that every implementation multiplies exactly the same logical inputs [1]. The files can be regenerated with `data/generate.py`.

3.3 Environment

Item	Value
CPU	Intel Core i7-12650H: 6 performance cores (P-cores, Hyper-Threading) and 4 efficiency cores (E-cores); 10 physical cores, 16 logical processors
Caches (P-core)	L1d 48 KB and L2 1.25 MB per core; L3 24 MB shared; 64-byte lines (reported by Windows [12])
RAM	15.7 GiB
OS	Windows 11 (build 10.0.26200)
C	gcc 16.1.0 (MSYS2 UCRT64), flags `-O2 -std=c11 -Wall -Wextra`
Java	Oracle JDK 21.0.10 (HotSpot), compiled with `javac --release 21`, run with `-Xms4g -Xmx4g -XX:+UseSerialGC`
Python	CPython 3.13.7
Measurement tools	psutil 7.2.2, NumPy 2.5.4 (reference only)

The i7-12650H is a hybrid processor. If the operating system moved a benchmark process from a P-core to a slower E-core during a measurement, the timings would mix two kinds of core. The P-core logical processors were identified with the Windows `GetSystemCpuSetInformation` API (EfficiencyClass 1: logical processors 0–11; EfficiencyClass 0: logical processors 12–15) [9]. Every benchmark process was pinned to processors 0–11 and given the HIGH priority class. All implementations are single-threaded. The laptop was connected to mains power during the runs, and heavy applications were closed.

All settings above were kept fixed for every configuration of the main run.

3.4 Measurement protocol

- **Timing boundary.** Only the multiplication kernel is timed, including allocation and zero-initialisation of the output matrix. Loading the input files happens before the timed region.
- **Timers.** Monotonic high-resolution timers were used: `QueryPerformanceCounter` in C (10 MHz, i.e. 100 ns resolution on this machine) [8], `System.nanoTime` in Java, and `time.perf_counter_ns` in Python [6].
- **Process structure.** Each (language, n) configuration ran as one process that loaded the inputs, performed the warm-up runs and then five measured repetitions. Warm-up runs were excluded from the samples: 5 for Java (to let the JIT compiler optimise the kernel) and 1 for C and Python (to warm caches and memory pages).
- **Result consumption.** After each multiplication, the sum of all elements of C (checksum) was computed outside the timed region and reported, so the computation cannot be eliminated.
- **Memory metric.** Peak working set (`peak_wset`, the peak amount of physical memory used by the process) was read with psutil [7] by an external harness for all three languages. Because Windows only exposes this value while the process exists, each benchmark process prints its results and then waits on standard input; the harness reads the peak working set and then releases the process. This metric covers the whole process: runtime (JVM or interpreter), loaded inputs and output.
- **Statistics.** Every raw measurement was stored in CSV. For each configuration the median and the interquartile range (IQR = Q3 − Q1) of the five repetitions are reported.

3.5 Correctness

Before benchmarking, each implementation passed unit tests in its own language: multiplication by the identity matrix, by a zero matrix (on both sides), a known 2 × 2 case, a known 3 × 3 case with non-integer values, and the checksum function. In addition, the products of all three implementations for n = 64 and n = 128 were compared with NumPy's `A @ B` using `numpy.allclose` with `rtol = 1e-9` and `atol = 1e-12`. The tolerance allows for differences in the order of floating-point additions between our loop and NumPy's optimised library, while remaining far stricter than any real error. The maximum absolute error was 5.3×10^{-15} and the maximum relative error 1.9×10^{-12} (n = 128 and n = 64, respectively).

3.6 Sizes and budget

The declared size sequence was n = 64, 128, 256, 512, 1024, 2048 and 4096. Each process had a budget of 900 s, covering warm-ups and the five repetitions. A configuration that exceeded the budget was killed and recorded as a timeout with its limit, and larger sizes for that language were skipped, since doubling n multiplies the work by eight. Memory was not a limiting factor: three float64 matrices at n = 4096 need about 400 MB.

4. Results

4.1 Runtime

Table 1. Kernel runtime: median and IQR of 5 repetitions (seconds). Timeouts at 900 s per process.

n	C median	C IQR	Java median	Java IQR	Python median	Python IQR
64	0.0000696	0.0000012	0.000448	0.000029	0.00781	0.00002
128	0.00111	0.00002	0.00151	0.00001	0.0613	0.0002
256	0.00837	0.00004	0.00941	0.00016	0.546	0.004
512	0.270	0.003	0.277	0.011	9.60	0.94
1024	2.18	0.04	3.78	0.23	113.6	11.3
2048	114.5	5.5	timeout	—	timeout	—
4096	timeout	—	skipped	—	skipped	—

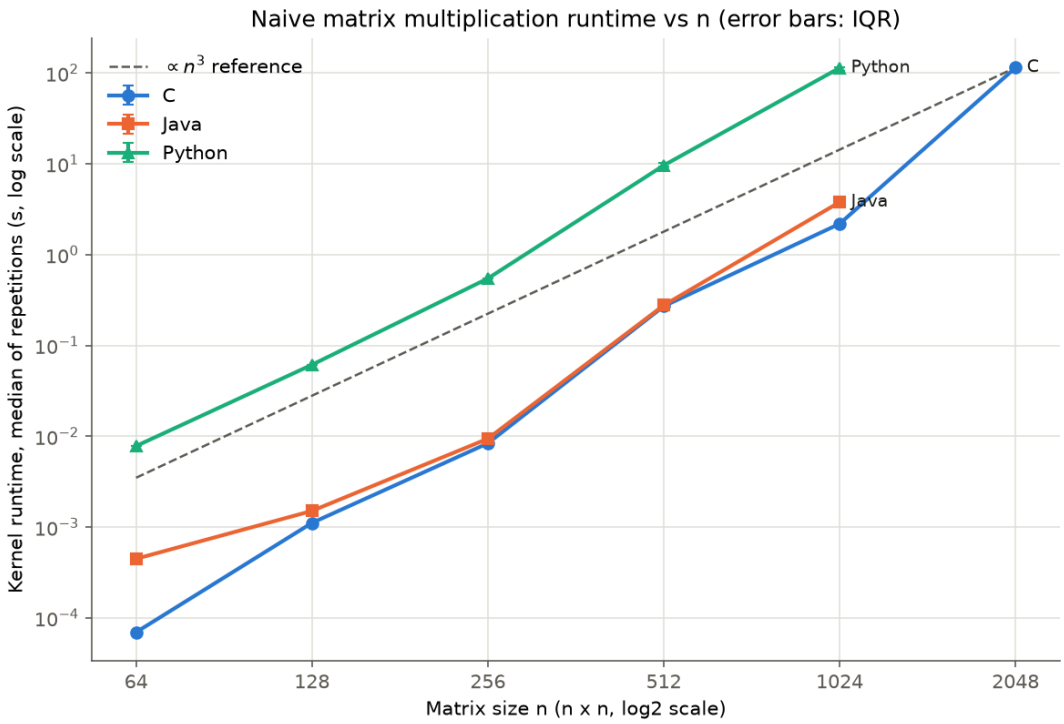


Figure 1. Kernel runtime against n (log-log scale), with error bars showing the IQR and a dashed n^3 reference line.

The largest feasible tested size was n = 2048 for C and n = 1024 for Java and Python. The largest size attempted was 4096 for C and 2048 for Java and Python.

Table 2. Growth ratio T(n)/T(n/2) of the median runtime. A purely cubic algorithm gives 8.

n	C	Java	Python
128	16.0	3.4	7.8
256	7.5	6.2	8.9
512	32.3	29.4	17.6
1024	8.1	13.7	11.8
2048	52.4	—	—

Table 3. Time per inner-loop iteration, T/n^3 (nanoseconds). A purely cubic cost gives a constant value. The last column gives the size of B ($8n^2$ bytes), the operand read with stride n.

n	C	Java	Python	Size of B
64	0.27	1.71	29.8	32 KB
128	0.53	0.72	29.2	128 KB
256	0.50	0.56	32.5	512 KB
512	2.01	2.06	71.6	2 MB
1024	2.03	3.52	105.8	8 MB
2048	13.33	—	—	32 MB

Variability was low: the IQR was below 2.5 % of the median for every C configuration up to n = 1024, and at most 6.4 % for Java. It increased for long runs: 4.8 % for C at n = 2048 and about 10 % for Python at n = 512 and 1024.

4.2 Memory

Table 4. Peak working set of the whole process (MiB).

n	C	Java	Python
64	3.9	46.8	13.3
128	4.2	48.4	14.2
256	5.3	53.4	20.5
512	9.7	71.8	42.8
1024	27.7	142.9	133.6
2048	99.8	—	—

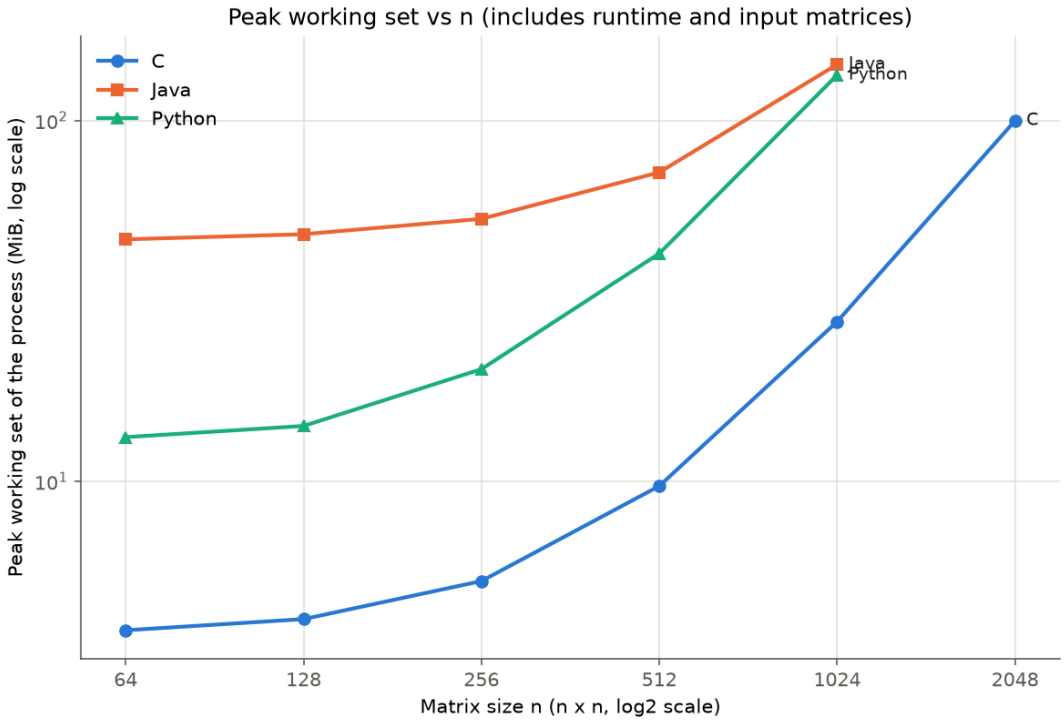


Figure 2. Peak working set against n.

4.3 Supplementary experiment: Java at n = 2048 with one warm-up

While analysing the main run, we found a bias in the budget design. Because the 900 s limit covered the whole process, Java (5 warm-ups + 5 repetitions = 10 multiplications) had a stricter effective limit than C and Python (1 + 5 = 6 multiplications). The Java timeout at n = 2048 could therefore reflect the number of warm-ups rather than the cost of one multiplication.

To test this, Java at n = 2048 was run again separately, with the same inputs, flags and pinning, one warm-up run (the same number as C) and a 2700 s budget. This run took place later on the same day as

the main run (2026-10-14, 21:45), so its thermal and background conditions may differ. Its results are reported separately and are not merged into Tables 1–4.

Table 5. Supplementary run compared with C at n = 2048 from the main run.

	C (main run)	Java (supplementary)
Warm-ups	1	1
Median (s)	114.5	129.2
IQR (s)	5.5	9.8
Repetitions (s)	121.4, 112.3, 108.6, 114.5, 117.9	119.8, 124.7, 138.7, 134.4, 129.2
Peak working set (MiB)	99.8	303.3
Checksum	−23415.104678236432	−23415.104678236432

The whole Java process took about 13 minutes, so with one warm-up it would have fitted the original 900 s budget. With five warm-ups (about 10 × 129 s ≈ 21 min) it would not. This confirms that the main-run timeout was caused by the budget design.

5. Discussion

Growth compared with the cubic model (H1). The runtime did not grow by a constant factor of 8. When n doubled and the strided operand B stayed within the same cache level, the ratio was close to 8 (C: 7.5 from 128 to 256 and 8.1 from 512 to 1024). When B crossed into the next level, the ratio jumped (C: 16 at 64→128, 32.3 at 256→512 and 52.4 at 1024→2048). Table 3 shows this more clearly. In C the time per iteration forms steps of 0.27 ns, about 0.5 ns, about 2.0 ns and 13.3 ns. These steps match B fitting in L1 (32 KB < 48 KB), L2 (128–512 KB < 1.25 MB), L3 (2–8 MB < 24 MB), and finally exceeding L3 (32 MB). This explanation is consistent with the data, but it is not proven, because cache misses were not measured with hardware counters. A complementary explanation is that power-of-two strides map many elements of a column of B to the same cache sets, causing conflict misses. Because every tested size was a power of two, the two effects cannot be separated here; comparing, for example, n = 1000 with n = 1024 could test this. In summary, the algorithm performs n^3 operations, but the cost per operation depends on where the data is stored. H1 holds only within a cache level.

Execution models (H2). H2 is supported. CPython was between 36 and 112 times slower than C (Table 1). The ratio was largest at small sizes (112 at n = 64), where C ran from the fastest cache level. Each Python iteration costs about 30 ns even when the data fits in cache, because the interpreter dispatches bytecode, checks types and creates a new float object for each intermediate result. Java was close to C at n = 256 and n = 512 (1.12 and 1.02 times the C runtime), which is consistent with HotSpot compiling the kernel to native code. Java was slower at n = 64 (1.71 ns per iteration against 0.27 ns). A plausible cause is that five warm-ups of a very small problem may not trigger full optimisation, or that fixed per-call costs dominate at that size; this was not investigated. At n = 1024, Java needed 1.73 times the C runtime. Possible explanations, such as array bounds checks or differences in the generated code, were not tested. In the supplementary run the gap narrowed to 1.13 at n = 2048. This is consistent with both implementations being limited by main-memory access at that size, which reduces the weight of language-specific differences.

Python and the memory hierarchy (H3). H3 is only partly supported. As predicted, the growth ratio stayed close to 8 for small sizes (7.8 and 8.9). However, it reached 17.6 from 256 to 512, and the time

per iteration rose from about 30 ns to 106 ns. Memory effects therefore matter in Python too, although less than in C. A plausible reason is that a list of lists stores about 32 bytes per element (a pointer plus a float object) instead of 8, and the float objects are scattered across the heap. B therefore exceeds each cache level at smaller n than in C, and its accesses are less predictable.

Memory. The peak working set has two components: a fixed runtime baseline and the matrices. C's baseline is about 4 MiB, and at n = 2048 its 99.8 MiB is close to the 96 MiB of three 2048 × 2048 float64 matrices. Java has a baseline of about 47 MiB (the JVM itself). Python starts at about 13 MiB but grows faster than C with n, which is consistent with its per-element object overhead. The Java peak at n = 2048 (303 MiB) is about three times the size of the matrices. This may include temporary buffers used while loading the input files, but the cause was not measured. Because the metric covers the whole process, it should not be read as "memory used by the multiplication" alone.

Variability. Relative variability was small for short runs and grew for long ones, reaching 10 % for Python at n = 1024. In the supplementary run the Java times rose across repetitions (from 120 s to 139 s). Insufficient warm-up would make times fall, not rise, so a more plausible cause is thermal throttling of the laptop during long runs. Processor frequency and temperature were not recorded, so this remains a hypothesis.

Limitations.

- All results come from one machine and operating system. They should not be generalised as "language X is faster than language Y".
- Cache behaviour was inferred from timings, not measured with hardware performance counters.
- Only powers of two were tested, which may amplify cache-conflict effects.
- The peak working set includes runtime and input data, and its baseline differs between languages.
- The C timer resolution is 100 ns, which is negligible compared with the shortest median (69.6 µs).
- The budget design penalised Java at n = 2048 (Section 4.3). The supplementary run used a different budget and number of warm-ups, and was executed at a different time.
- Before the runs, the harness did not record whether CPU pinning and priority were applied successfully, so this cannot be confirmed for the reported data. The harness was corrected afterwards to log any such failure.
- CPU frequency and temperature were not recorded.

6. Conclusions

Within the tested scope (one Intel Core i7-12650H laptop, Windows 11, the stated compilers and runtimes, float64 inputs, n from 64 to 2048), the study establishes that:

1. The runtime of the naive i-j-k multiplication follows the cubic model only while the strided operand stays within one level of the memory hierarchy. Each time it moves to a slower level, the time per iteration rises by a factor of about 2 to 7, so the memory layout dominates the deviation from n^3.
2. CPython was 36–112 times slower than C for the same algorithm, mainly because of per-operation interpreter overhead. Its time per iteration also increased with n, so memory effects are not absent.
3. After JIT warm-up, Java performed close to C for mid-range sizes. It was 1.7 times slower at n = 1024 and 1.13 times slower at n = 2048, where both appear to be memory-bound.

4. The peak working set grows with n^2 on top of a runtime-specific baseline, which is largest for the JVM.
5. Timeout budgets that include warm-ups can bias comparisons between languages with different warm-up requirements.

The study does not establish the exact cause of the per-level slowdowns (capacity misses or set conflicts), the reason for the Java–C gap at n = 1024, or the source of the extra Java memory. Hardware performance counters, non-power-of-two sizes and frequency logging would address these questions.

7. References

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8. Reproducibility materials

All materials are in the repository above, in `assignment-1/`, at tag `assignment-1`:

- Source code: `c/`, `java/`, `python/`; benchmark harness `benchmark.py`; tests `run_tests.py`; NumPy cross-check `cross_check.py`.
- Input generation: `data/generate.py` (seeds and parameters in `data/README.md`).
- Configuration: `bench_config.json` (main run) and `bench_config_supplementary.json` (Section 4.3).
- Raw data: `results/raw_measurements.csv` and `results/supplementary/raw_measurements.csv`; environment metadata in `environment.json`.
- Analysis: `results/analyze.py` produces the summary tables and figures.
- Exact setup, build, test, run and analysis commands: `assignment-1/README.md`.

Generative AI disclosure

Claude (Anthropic), used through Claude Code, assisted with this assignment. It proposed the experimental design, which the author reviewed and approved. It implemented the benchmark code (the three implementations, the tests, the harness and the analysis scripts) following that design. It explained the code and the results, and it drafted the text of this report from the measurements and the analysis discussed with the author. The author ran the benchmarks on their own machine, identified the warm-up/budget bias and decided to run the supplementary experiment, reviewed and edited the report, and checked the measurements and references. The author is responsible for the content and can explain the code and the conclusions. All numerical results come from the measurements in the repository; none were generated by the AI tool.